

HOW MULTITASKING AFFECTS PRODUCTIVITY AND BRAIN HEALTH

Focus, Attention and the Science Behind Doing Too Much at Once • LEVEL B



Multitasking has become a common habit in modern life — but research suggests our brains are far less capable of it than we believe.

Multitasking seems like a great way to get a lot done at once, but research has shown that our brains are not nearly as good at handling multiple tasks as we like to think. In fact, some studies suggest that multitasking can actually hamper your productivity by reducing your comprehension, attention, and overall performance. What makes multitasking such a productivity killer? It might seem like you are accomplishing multiple things at the same time, but what you are really doing is rapidly shifting your attention and focus from one thing to the next. This constant switching makes it difficult to tune out distractions and can cause mental blocks that slow you down.

"Multitasking refers to working on two or more tasks simultaneously, switching back and forth between them, or performing a number of tasks in rapid succession — often at the cost of focus and quality."

— Definition

- How Multitasking Hampers Productivity
- Brain Function in Multitaskers
- Teens, Media and Multitasking

How Multitasking Hampers Productivity

Multitasking takes a serious toll on **productivity**. Our brains lack the ability to truly perform multiple tasks at the same time — in moments where we think we are multitasking, we are most likely just switching quickly from one task to another. Focusing on a single task is a far more effective approach, for several important reasons.

Multitasking Is Distracting

Multitaskers often report feeling more distracted than people who focus on one task at a time. This makes sense when you consider that, by habit, multitaskers are constantly switching to a new task, effectively making themselves more **distractible**. Some research suggests that this habit of **distraction** can carry over even when a person is not actively **multitasking** — making it harder to maintain focus in general.

Multitasking Slows You Down

Although it may seem contrary to popular belief, we tend to work more slowly and less efficiently when we multitask. Psychologists refer to this as '**task switch costs**' — the negative effects that result from switching back and forth between tasks. These costs occur because each switch increases the mental demand placed on the brain. Furthermore, switching tasks prevents us from working on **autopilot** — that **automatic**, efficient mode we enter when we focus on a familiar task without interruption. When we keep switching, we lose access to this mental shortcut and end up working more slowly as a result.

Multitasking Impairs Executive Function

Multitasking is managed by what researchers call **executive function** — the brain's system for controlling and organising **cognitive** processes. There are two key stages to this process: first, goal shifting (deciding to switch from one task to another), and second, rule activation (adjusting your mental approach to suit the new task). While each switch may only take a fraction of a second, these small delays add up quickly when you are switching repeatedly. In situations where safety or **efficiency** matters — such as driving in heavy traffic — even tiny lapses in attention can have serious consequences.

Multitaskers Make More Mistakes

Multitasking may also **impair** accuracy and make you more prone to errors. Research has shown that students who multitask during class tend to have lower academic grades, and if they continue at home, they often take longer to complete their homework. Adults are not immune either — one study found that older adults were significantly more likely to make mistakes while driving if they were **simultaneously** engaged in another task.

■ SINGLE-TASKING

- Allows the brain to work on **autopilot** for familiar tasks, saving mental energy.
- Reduces **distraction** and improves the ability to **tune out irrelevant** information.
- Leads to fewer **cognitive errors** and higher quality output.
- Allows deeper focus, which improves comprehension and retention.
- Results in greater **efficiency** and faster task completion overall.

■ MULTITASKING

- Increases **task switch costs**, slowing overall productivity.
- Makes people more **distractible**, even when not multitasking.
- Impairs **executive function** and leads to more mistakes.
- Associated with elevated stress and reduced **cognitive** performance.
- May have a particularly **detrimental** effect on **adolescent** brain development.

Brain Function in Multitaskers

Doing several different things at once can **impair cognitive** ability — even in people who multitask frequently. In fact, research suggests that people tend to **overestimate** their ability to multitask, and those who do it most often are frequently among the least effective at it. **Chronic** multitaskers tend to show higher levels of **impulsivity** than their peers, and they are often more willing to downplay the risks of juggling multiple tasks.

This happens partly because the brain has limited **cognitive** resources. Several networks in the brain work together to guide our behaviour when we set out to complete a task — including setting a goal, identifying the information needed to achieve it, and disregarding **irrelevant** distractions. When we try to run this process for multiple tasks at once, it can lead to **cognitive errors**, such as failing to filter out irrelevant information and becoming more easily distracted. Whether **chronic multitasking** causes these problems, or whether people with these traits are simply more drawn to multitasking, remains an open question in the research.

40%

drop in productivity caused by multitasking

2.5x

longer to complete tasks when constantly switching

~0.4s

lost per switch — adds up to minutes every hour

Teens, Media and Multitasking

The negative effects of **chronic multitasking** may be especially **detrimental** for **adolescent** minds. During the teenage years, the brain is still forming important **neural connections**. Constantly splitting attention and being pulled in different directions by multiple streams of information could have a lasting, negative impact on how those connections develop.

Interestingly, research on media **multitasking** — using more than one form of technology or media at the same time — has produced some mixed results. One study found that people who frequently engage in media multitasking may be more **adept** at integrating information from different **sensory** sources, such as sound and vision. However, the same group performed worse than lighter multitaskers when only one source of sensory information was available. In other words, heavy media multitaskers may develop a kind of broad awareness at the cost of deeper, more focused attention — a trade-off that is not always beneficial.

How to Break the Multitasking Habit

If **multitasking** is negatively affecting your **productivity** or wellbeing, there are practical steps you can take to change your habits. Start by making an honest **assessment** of the tasks you are trying to accomplish at any given moment, then decide which one deserves your full attention first. Small changes in how you organise your work can make a significant difference to your **efficiency** and focus.

1 Focus on one task at a time

Limit yourself to a single task whenever possible. If you must combine two, pair something **automatic** — like tidying up — with something that requires more active thinking.

2 Use the 20-minute rule

Instead of constantly switching between tasks, fully **devote** your attention to one thing for 20 minutes before moving on. This reduces **task switch costs** and helps you enter a state of deeper focus.

3 Batch your tasks

Group similar tasks together and schedule a set time to handle them — for example, checking emails only twice a day. Batching tasks frees your mind to concentrate on one thing at a time without constant interruption.

4 Limit distractions

Find a quiet place to work, switch your phone to silent, and turn off notifications. Reducing environmental distractions is one of the simplest and most effective ways to improve your focus and **productivity**.

5 Practise mindfulness

Adding **mindfulness** to your daily routine can help you become more aware of when you are **multitasking**. With practice, mindfulness can also improve your ability to sustain attention and resist the urge to switch tasks unnecessarily.

ACTIVITIES

1 Read the article and decide if these statements are True (T), False (F), or Not Given (NG).

- 1 Multitasking means doing several things at exactly the same time.
- 2 Switching between tasks increases the mental demand placed on the brain.
- 3 Students who multitask always receive failing grades.
- 4 Chronic multitaskers tend to overestimate their ability to handle multiple tasks.
- 5 Media multitasking has been shown to have no benefits at all.
- 6 The 20-minute rule involves focusing on one task for 20 minutes before switching.

2

Match the highlighted words and phrases in the article to their meanings.

- 1 to make something less effective or weaker
- 2 something that is not relevant or useful to the current situation
- 3 the brain's automatic, efficient way of completing a familiar task
- 4 the ability to combine information from more than one source
- 5 the tendency to act quickly without thinking carefully
- 6 particularly harmful, especially over a long period of time

3

Read the article again and answer these questions in full sentences.

- 1 What are 'task switch costs' and why do they slow us down?
- 2 Why might multitasking be especially harmful for teenagers?
- 3 What did the media multitasking study reveal about heavy multitaskers?
- 4 What is the purpose of 'batching' tasks, according to the article?
- 5 Why does focusing on a single task allow us to work on 'autopilot'?

4

Think about your own study or work habits. Write a short paragraph (80–100 words) describing how multitasking affects you and what changes you could make.

Use ideas and vocabulary from the article.

Consider: When do you multitask? Does it help or harm you? Which tip would you try first?

■ TALKING POINTS

Do you think multitasking is a skill or a bad habit? Can it ever be useful?

Why do you think so many people believe they are good at multitasking?

Should schools teach students how to focus? How might they do this?

How has technology made multitasking more common — and more difficult to avoid?